

Central Heating Water Reroute Roadmap

A 10-Stage Surface-Mounted Two-Pipe Copper System Installation Guide for Leasehold Properties

Stage 1: Pre-Work Approvals & Administrative Safeguards

Before physical execution begins, formal written consent from freeholders and insurer notifications are mandatory to mitigate liability.

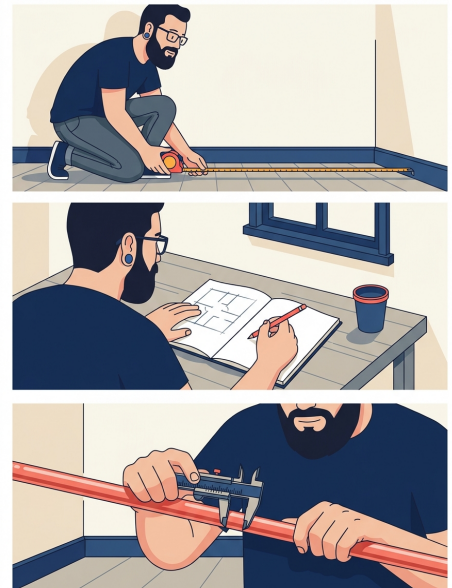
- Verify lease agreement covenants for alterable structures.
- Validate system layout and safety boundaries with Gas Safe registered engineer.
- Establish site log for helper verification and quality assurance.



Stage 2: Route Measurement & Physical Layout Mapping

Metrology and physical layout drafting ensure precise pipe routing along room perimeters before purchasing supplies.

- Measure physical walls directly rather than relying solely on floorplans.
- Calculate double the perimeter distance for flow and return lines + 10% waste buffer.
- Verify outer pipe diameter with metal vernier calipers for accurate clip sizing.



Stage 3: System Drainage & Electrical Isolation

System drainage and power isolation prevent electrical hazards and localized flooding during line cutting.

- Switch off fused spur electrical connection and confirm zero power.
- Connect garden hose to brass drain valve and evacuate sludge into catch bucket.
- Bleed panel radiators to relieve system air locks and residual pressure.



Stage 4: Route Preparation & Wall Penetrations

Structural wall penetrations require electronic scanning to prevent accidental contact with hidden wiring or utilities.

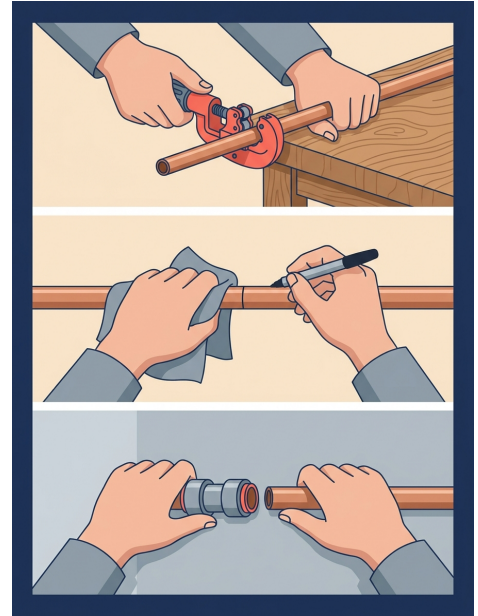
- Scan wall surfaces with electronic stud/metal detectors before drilling.
- Drill horizontal penetrations cleanly just above floorboard skirting levels.
- Install rubber-lined acoustic clips to support exactly TWO copper pipe lines.



Stage 5: Pipe Preparation & Push-Fit Golden Sequence

The 5-step pipe prep protocol ensures watertight integrity and seal protection for push-fit fittings.

- 1. Cut square with rotary cutter; 2. Deburr inner/outer edges thoroughly.
- 3. Clean copper surface to a shine; 4. Mark insertion depth clearly.
- 5. Push pipe firmly into fitting until fully seated against internal stop.



Stage 6: Radiator Branchwork & Parallel Connections

Radiator branches connect via parallel T-junctions off the two main trunk lines to ensure independent heat distribution.

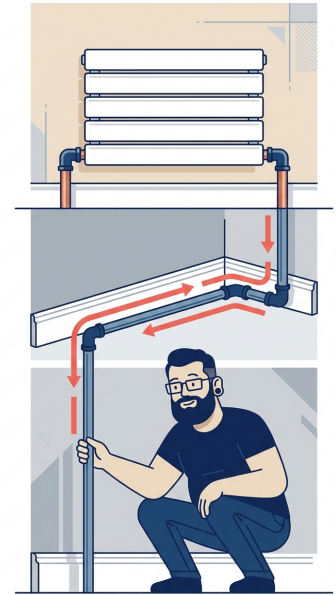
- Splice brass T-fittings into top (flow) and bottom (return) copper trunks.
- Branch short vertical copper pipes upward directly beneath radiator valves.
- Connect into radiator valves using spanners on classic grooved panel units.



Stage 7: Final Loop Closure & Continuous Circuit Routing

Completing the loop requires terminating terminal radiators with 90-degree elbows and routing continuous return lines.

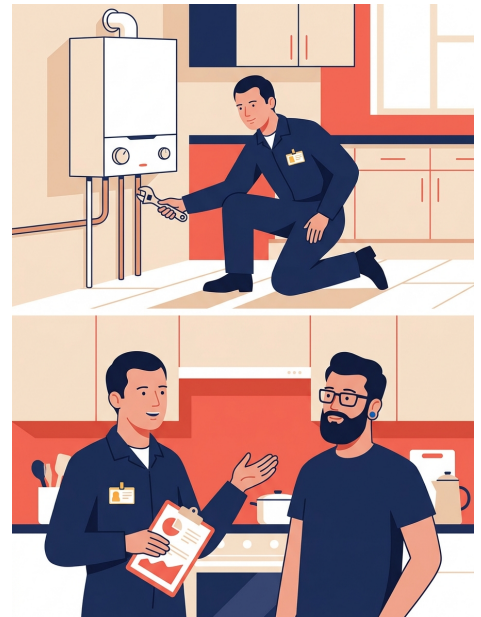
- Use 90-degree elbow fittings for terminal radiator connections.
- Maintain continuous blue-tinted return pipe route back toward boiler location.
- Hand-trace the physical circuit to verify zero crossover or unclosed lines.



Stage 8: Gas Safe Reconnection & Professional Handover

All boiler underslung connections and gas appliances must be handled strictly by a certified Gas Safe engineer.

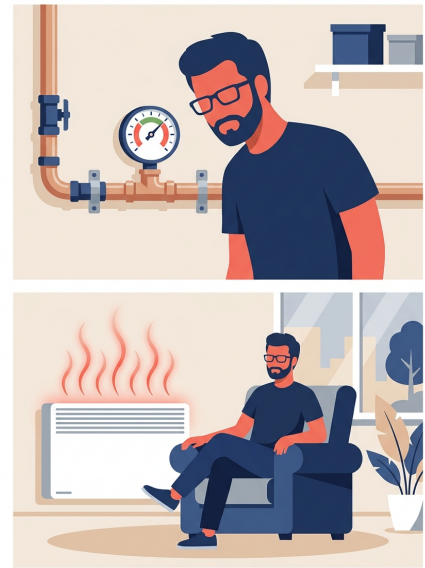
- Clean-shaven Gas Safe engineer connects flow and return lines under boiler.
- Homeowner observes handover attentively from a safe distance.
- Formal certification and safety clearance issued upon connection approval.



Stage 9: System Refill, Bleeding & Hydrostatic Testing

Refilling and bleeding the hydronic loop confirms joint integrity under operational pressure.

- Refill system to stable operating pressure between 1.0 and 1.5 bar.
- Inspect every joint with dry blue paper tissue to detect micro-moisture.
- Verify heat emission from panel radiators with comfortable warm convection.



Stage 10: Aesthetic Boxing-In & Permanent Access

Concealing pipework behind painted timber boxing enhances aesthetics while preserving access for future maintenance.

- Fix wooden timber battens above and below exactly TWO copper pipe lines.
- Fasten painted grey MDF panels over battens to enclose the run.
- Install hinged access hatches over isolation valves for maintenance access.

